Sheet 6

x	-2	-1	0	1	2
$f(x)$	$1/8$	$2/8$	$2/8$	$2/8$	$1/8$

a) Verify that the given functions are probability mass functions.

b) determine the cumulative distribution function for the random variable in the given table. (Find a formula for F(x).)

c) Determine the mean and variance of the random variable In the given table

Solution

a) 1/8+2/8+2/8+2/8+1/8=1, then f(x) is a probability mass function.

b)

$$F(x) = \begin{cases} 0, & x < -2 \\ 1/8 & -2 \le x < -1 \\ 3/8 & -1 \le x < 0 \\ 5/8 & 0 \le x < 1 \\ 7/8 & 1 \le x < 2 \\ 1 & 2 \le x \end{cases} \quad \text{where} \quad \begin{array}{l} f_X(-2) = 1/8 \\ f_X(-1) = 2/8 \\ f_X(0) = 2/8 \\ f_X(1) = 2/8 \\ f_X(2) = 1/8 \end{array}$$

Mean and Variance

$$\mu = E(X) = 0f(0) + 1f(1) + 2f(2) + 3f(3) + 4f(4)$$

$$= 0(0.2) + 1(0.2) + 2(0.2) + 3(0.2) + 4(0.2) = 2$$

$$V(X) = 0^2 f(0) + 1^2 f(1) + 2^2 f(2) + 3^2 f(3) + 4^2 f(4) - \mu^2$$

$$= 0(0.2) + 1(0.2) + 4(0.2) + 9(0.2) + 16(0.2) - 2^2 = 2$$

Verify that the following functions are probability mass functions, and determine the requested probabilities.

x	-2	-1	0	1	2
$f(x)$	1/8	2/8	2/8	2/8	1/8

(a) $P(X \leq 2)$ (b) $P(X > -2)$
(c) $P(-1 \leq X \leq 1)$ (d) $P(X \leq -1 \text{ or } X = 2)$

Solution

All probabilities are greater than or equal to zero and sum to one, then f(x) is Prob. Mass function.

a) $P(X \leq 2) = 1/8 + 2/8 + 2/8 + 2/8 + 1/8 = 1$
b) $P(X > -2) = 2/8 + 2/8 + 2/8 + 1/8 = 7/8$
c) $P(-1 \leq X \leq 1) = 2/8 + 2/8 + 2/8 = 6/8 = 3/4$
d) $P(X \leq -1 \text{ or } X=2) = 1/8 + 2/8 + 1/8 = 4/8 = 1/2$

Verify that the following functions are probability mass functions, and determine the requested probabilities.

$$f(x) = \frac{2x + 1}{25}, \quad x = 0, 1, 2, 3, 4$$

(a) $P(X = 4)$ (b) $P(X \leq 1)$
(c) $P(2 \leq X < 4)$ (d) $P(X > -10)$

x	0	1	2	3	4
P(x)	1/ 25	3/25	1/5	7/25	9/25

Probabilities are nonnegative and sum to one.

a) $P(X = 4) = 9/25$

b) $P(X \leq 1) = 1/25 + 3/25 = 4/25$

c) $P(2 \leq X < 4) = 5/25 + 7/25 = 12/25$

d) $P(X > -10) = 1$

A discrete random variable, X, has probability mass function

$$x = -2, -1,\ 0,\ 1,\ 2. \quad p(x) = c(x-3)^2,$$

(a) Find the value of the constant c.

(b) Find the mean and variance of X.

a) Science P(x) is Probability mass function, then P(-2)+P(-1)+P(0)+P(1)+P(2)=1
25c+16c+9c+4c+c=1, then c=1/55.

x	-2	-1	0	1	2
P(x)	25C	16C	9C	4C	C

b) E(x)= -2 P(-2)-1P(-1) +0 P(0) +1 P(1)+2 P(2)=
c) V(x)= $(-2)^2P(-2)$+ $(-1)^2P(-1)$+ $(0)^2P(0)$+ $(1)^2P(1)$+ $(2)^2P(2) =$

3-22. In a semiconductor manufacturing process, three wafers from a lot are tested. Each wafer is classified as *pass* or *fail*. Assume that the probability that a wafer passes the test is 0.8 and that wafers are independent. Determine the probability mass function of the number of wafers from a lot that pass the test.

Solution

X = number of wafers that pass

$P(X=0) = (0.2)^3 = 0.008$

$P(X=1) = 3(0.2)^2(0.8) = 0.096$

$P(X=2) = 3(0.2)(0.8)^2 = 0.384$

$P(X=3) = (0.8)^3 = 0.512$

x	0	1	2	3
f(x)	0.008	0,096	0.348	0.512

3-24. An assembly consists of two mechanical components. Suppose that the probabilities that the first and second components meet specifications are 0.95 and 0.98. Assume that the components are independent. Determine the probability mass function of the number of components in the assembly that meet specifications.

Solution

X = number of components that meet specifications

$P(X=0) = (0.05)(0.02) = 0.001$

$P(X=1) = (0.05)(0.98) + (0.95)(0.02) = 0.068$

$P(X=2) = (0.95)(0.98) = 0.931$

x	0	1	2
f(x)	0.001	0.068	0.931

5- Suppose X is a random variable with $E(X) = 4$ and $V(X) = 9$. Let $Y = 4X + 5$. Compute $E(Y)$ and $V(Y)$.

Solution

$$E(Y) = E(4X + 5) = 4E(x) + 5 = 16 + 5 = 21$$

$$V(x) = V(4X + 5) = 16V(X) = 16(9) = 144$$

Prove that if X is a discrete random variable then for any constants a and b we have $V(aX + b) = a^2 V(X)$

Solution

$V(Y) = V(aX + b) =$

$= E((aX + b)^2) - (E(aX + b))^2 =$

$=E(a^2X^2 + 2abX + b^2) - (aE(X) + b)^2 =$

$= a^2E(X^2) + 2abE(X) + b^2 - a^2E(X)^2 - 2abE(X) - b^2 =$

$= a^2E(X^2) - a^2E(X)^2 =$

$= a^2V(X)$

x	0	1	2	3	4
f(x)	0	1/30	4/30	9/30	16/30

If f(x) is a prob. Mass function, then find E(x) and E(x(x-1))

Sol.

$$E(x) = \sum xf(x) = 3.33$$

$$E(x) = \sum x(x-1)f(x) = E(x^2 - x)f(x) = 8.47$$

x	0	1	2	3
f(x)	1/30	3/10	1/2	1/6

$$E(x) = \sum xf(x) = 0.9666$$

A continuous random variable has a pdf of X is given by

$$f(x) = \begin{cases} a + bx^2 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Suppose also that you are told that E(X) = 3

(a) Find a and b.

(b) Determine the CDF, F(x).

Solution

$$\int_0^1 (a + bx^2)dx = 1$$

$$[ax + \frac{bx^3}{3}]_0^1 = 1$$

$a + \frac{b}{3} = 1$---------------------(1)

$$E(X) = \int_0^1 x\, f(x)dx = 3$$

$$\int_0^1 x\,(a + bx^2)dx = [\mathrm{a}\frac{x^2}{2} + \mathrm{b}\frac{x^4}{4}]_0^1 = 3$$

$$\frac{a}{2} + \frac{b}{3} = 3 \text{-------------------------------} (2)$$

By solving 1&2 then a=, b=

A ,9-=b=30

$$F(x) = \int_{-\infty}^{x} f(x)dx$$

$$For - \infty < x < 0, F(x) = \int_{-\infty}^{x} 0\ dx = 0$$

$$For 0 < x < 1, F(x) = \int_{-\infty}^{0} 0\ dx + \int_{0}^{1} (-9 + 30x^2)dx + \int_{0}^{x} 0dx = -9x + 3x^3$$

$$For 1 < x < \infty, \qquad F(x) = \int_{-\infty}^{0} 0\ dx + \int_{0}^{x} (-9 + 30x^2)dx = 1$$

$$F(x) = \begin{cases} 0, & x < 0 \\ -9x + 3x^3 & 0 < x < 1 \\ 1, & 1 < x, \end{cases}$$

$$f(x) = \begin{cases} \frac{x}{8}, 3 < x < 5 \\ 0, O.W \end{cases}$$ is a pdf, then find The mean, the variance and also find:

(a) $P(X < 4)$

(b) $P(X > 3.5)$

(c) $P(4 < X < 5)$

(d) $P(X < 4.5)$

(e) $P(X < 3.5 \text{ or } X > 4.5)$

Solution

$$E(X) = \int_3^5 x \frac{x}{8} dx = \left. \frac{x^3}{24} \right|_3^5 = \frac{5^3 - 3^3}{24} = 4.083$$

$$V(X) = \int_3^5 (x - 4.083)^2 \frac{x}{8} dx = \int_3^5 \left(\frac{x^3}{8} - \frac{8.166x^2}{8} + \frac{16.6709x}{8} \right) dx$$

$$= \frac{1}{8} \left. \left(\frac{x^4}{4} - \frac{8.166x^3}{3} + \frac{16.6709x^2}{2} \right) \right|_3^5 = 0.3264$$

a) $P(X < 4) = \int_3^4 \frac{x}{8}dx = \frac{x^2}{16}\Big|_3^4 = \frac{4^2 - 3^2}{16} = 0.4375$, because $f_X(x) = 0$ for x < 3.

b) , $P(X > 3.5) = \int_{3.5}^5 \frac{x}{8}dx = \frac{x^2}{16}\Big|_{3.5}^5 = \frac{5^2 - 3.5^2}{16} = 0.7969$ because $f_X(x) = 0$ for x > 5.

c) $P(4 < X < 5) = \int_4^5 \frac{x}{8}dx = \frac{x^2}{16}\Big|_4^5 = \frac{5^2 - 4^2}{16} = 0.5625$

d) $P(X < 4.5) = \int_3^{4.5} \frac{x}{8}dx = \frac{x^2}{16}\Big|_3^{4.5} = \frac{4.5^2 - 3^2}{16} = 0.7031$

e) $P(X > 4.5) + P(X < 3.5) = \int_{4.5}^5 \frac{x}{8}dx + \int_3^{3.5} \frac{x}{8}dx = \frac{x^2}{16}\Big|_{4.5}^5 + \frac{x^2}{16}\Big|_3^{3.5} = \frac{5^2 - 4.5^2}{16} + \frac{3.5^2 - 3^2}{16} = 0.5$.

4-12. Suppose the cumulative distribution function of the random variable X is

$$F(x) = \begin{cases} 0 & x < -2 \\ 0.25x + 0.5 & -2 \le x < 2 \\ 1 & 2 \le x \end{cases}$$

Solution

a) $P(X < 1.8) = P(X \le 1.8) = F_X(1.8)$ because X is a continuous random variable.

Then, $F_X(1.8) = 0.25(1.8) + 0.5 = 0.95$

b) $P(X > -1.5) = 1 - P(X \le -1.5) = 1 - .125 = 0.875$

c) $P(X < -2) = 0$

d) $P(-1 < X < 1) = P(-1 < X \le 1) = F_X(1) - F_X(-1) = .75 - .25 = 0.50$

Binomial Distributions

X = the number of successes in the n trials in a Bernoulli Process

The random variable X has a binomial distribution with parameters n (number of trials) and p (probability of success), and we write:

$$X \sim \text{Binomial}\,(n, p) \quad \text{or} \quad X \sim b\,(x; n, p)$$

The probability distribution of X is given by:

$$f(x) = P(X = x) = b(x; n, p) = \begin{cases} \binom{n}{x} p^x \, (1-p)^{n-x} \; ; & x = 0, 1, 2, \ldots, n \\ 0 \; ; & \textit{otherwise} \end{cases}$$

Suppose that in a particular sheet of 100 postage stamps, 3 are defective. The inspection policy is to look at 5 randomly chosen stamps on a sheet and to release the sheet into circulation if none of those five is defective. Write down the random variable, the corresponding probability distribution and then determine the probability that the sheet described here will be allowed to go into circulation.

Solution.
Let X be the number of defective stamps in the sheet. Then X is a binomial random variable with probability distribution

$$P(X = x) = C(5, x)(0.03)^x(0.97)^{5-x},\ \ x = 0, 1, 2, 3, 4, 5.$$

$$P(\text{sheet goes into circulation}) = P(X = 0) = (0.97)^5 = 0.859$$

A student has no knowledge of the material to be tested on a true-false exam with 10 questions. So, the student flips a fair coin in order to determine the response to each question.
(a) What is the probability that the student answers at least six questions correctly?
(b) What is the probability that the student answers at most two questions correctly?

Solution.

(a) Let X be the number of correct responses. Then X is a binomial random variable with parameters $n = 10$ and $p = \frac{1}{2}$. So, the desired probability is

$$P(X \geq 6) = P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$$
$$= \sum_{x=6}^{10} C(10, x)(0.5)^x(0.5)^{10-x} \approx 0.3769.$$

(b) We have

$$P(X \leq 2) = \sum_{x=0}^{2} C(10, x)(0.5)^x(0.5)^{10-x} \approx 0.0547$$

The probability of hitting a target per shot is 0.2 and 10 shots are fired independently.
(a) What is the probability that the target is hit at least twice?
(b) What is the expected number of successful shots?

Solution.
Let X be the number of shots which hit the target. Then, X has a binomial distribution with $n = 10$ and $p = 0.2$.
(a) The event that the target is hit at least twice is $\{X \geq 2\}$. So,

$$\begin{aligned} P(X \geq 2) =& 1 - P(X < 2) = 1 - P(0) - P(1) \\ =& 1 - C(10,0)(0.2)^0(0.8)^{10} - C(10,1)(0.2)^1(0.8)^9 \\ \approx& 0.6242 \end{aligned}$$

(b) $E(X) = np = 10 \cdot (0.2) = 2.$

Let X be a binomial random variable with parameters $(12, 0.5)$. Find the variance and the standard deviation of X.

Solution.
We have $n = 12$ and $p = 0.5$. Thus, $\text{Var}(X) = np(1-p) = 6(1-0.5) = 3$.

The standard deviation is $SD(X) = \sqrt{3}$

An exam consists of 25 multiple choice questions in which there are five choices for each question. Suppose that you randomly pick an answer for each question. Let X denote the total number of correctly answered questions. Write an expression that represents each of the following probabilities.
(a) The probability that you get exactly 16, or 17, or 18 of the questions correct.
(b) The probability that you get at least one of the questions correct.

Solution.
(a) $P(X = 16 \text{ } or \text{ } X = 17 \text{ } or \text{ } X = 18) = C(25, 16)(0.2)^{16}(0.8)^{9} + C(25, 17)(0.2)^{17}(0.8)^{8} + C(25, 18)(0.2)^{18}(0.8)^{7}$

(b) $P(X \geq 1) = 1 - P(X = 0) = 1 - C(25, 0)(0.8)^{25}$

Suppose that 3% of computer chips produced by a certain machine are defective. The chips are put into packages of 20 chips for distribution to retailers. What is the probability that a randomly selected package of chips will contain at least 2 defective chips?

Solution

$$P(X \geq 2) = 1 - P(X < 2)$$
$$= 1 - C(20,0) \times 0.03^0 \times 0.97^{20} - C(20,1) \times 0.03^1 \times 0.97^{19}$$
$$=0.119838022$$

The probability of winning a lottery is 1/300. If you play the lottery 200 times, what is the probability that you win at least twice?

Solution

$$P(x \geq 2) = 1 - P(x < 2) = 1 - [P(x = 0) + P(x = 1)]$$

$$= 1 - [C(200,0) \times \left(\frac{1}{300}\right)^{0} \times \left(\frac{299}{300}\right)^{200} + C(200,0) \times \left(\frac{1}{300}\right)^{1} \times \left(\frac{299}{300}\right)^{199}$$